Solution Requirements

FitBuddy – AI Fitness Plan Generator using Gemini Models

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

FitBuddy is a web-based application that uses AI to generate personalized workout plans and nutrition tips based on a user's fitness goals. It uses FastAPI, Google Gemini models, SQLite with SQLAlchemy, and Jinja2 templates. The application accepts user details, generates a customized 7-day fitness plan, supports feedback-based plan updates, and provides an administrative view of users and plans.

• **Functional Requirements (FR)** define the specific behaviors, functions, and features FitBuddy must perform.

• **Non-Functional Requirements (NFR)** define criteria such as performance, usability, responsiveness, security, reliability, and maintainability.

Team Members and Work Allocation

S.No	Name / Role	Assigned Work
1	**Pavithra M** Team Leader	**Milestone 1 – Model Selection & Architecture** Research/select Gemini model, define application architecture, and set up the development environment.
2	**Manisha S** Team Member	**Milestone 2 – Core Functionalities** Develop workout-plan generation, nutrition/recovery tip generation, feedback-based plan updating, and user/plan storage.
3	**Monica G** Team Member	**Milestone 3 – routes.py Development** Implement the main routing logic, connect routes to Gemini functions and database operations, and handle user input.
4	**Monisha Anandan** Team Member	**Milestone 4 – Frontend Development** Design the HTML/CSS interface and create dynamic Jinja2 templates for input, results, feedback, and admin views.
5	**Mridula M** Team Member	**Milestone 5 – Deployment & Testing** Prepare local deployment, configure dependencies/environment variables, run the application, and verify local deployment.

Step 1: Functional Requirements (FR)

The following requirements are derived from the FitBuddy project workflow and documented features.

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**User Input & Profile**	Collect name, user ID, age, weight, fitness goal, and workout intensity through the web form.	High
2	**Workout Generation**	Generate a personalized 7-day, day-wise workout plan using Gemini 1.5 Pro based on user goal and intensity.	High
3	**Nutrition / Recovery Tips**	Generate a concise nutrition or recovery tip using Gemini Flash according to the selected fitness goal.	High
4	**Feedback & Plan Update**	Accept user feedback such as adding cardio or yoga and regenerate the existing plan using Gemini 1.5 Pro.	High
5	**Data Storage**	Store user information and generated plans in SQLite through SQLAlchemy, preserving original and updated plans.	High
6	**Admin View**	Provide /view-all-users to display registered users and their original and updated plans.	Medium
7	**API & Web Interface**	Provide FastAPI endpoints together with interactive HTML/Jinja2 pages for user interaction and API testing.	Medium

Step 2: Non-Functional Requirements (NFR)

These requirements describe how well the FitBuddy system should operate.

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Performance**	Gemini-based generation should provide practical responses suitable for real-time web interaction.	High
2	**Usability**	The interface should be clear and easy to use, with a focused input-to-result flow and readable result pages.	High
3	**Responsiveness**	Frontend layouts should support mobile screens using responsive CSS and media queries.	Medium
4	**Maintainability**	Use a modular FastAPI architecture separating routes, AI generation, database logic, templates, and static assets.	High
5	**Security**	Sensitive Gemini API credentials should be stored through environment variables rather than hard-coded in application code.	High
6	**Reliability**	The application should preserve original plans when feedback-based updates are made.	High
7	**Scalability**	The API-based Gemini integration and modular structure should support future expansion and deployment.	Medium

External Interfaces / Technology Stack

Interface / Component	FitBuddy Implementation
AI Interface	Google Gemini 1.5 Pro for workout generation and feedback updates; Gemini Flash for nutrition/recovery tips.
Backend	FastAPI with routes.py for routing and user-input processing.
Frontend	HTML, CSS and Jinja2 templates: index.html, result.html, all_users.html.
Database	SQLite with SQLAlchemy ORM for user and plan persistence.
Server / API Testing	Uvicorn local server and FastAPI interactive documentation at /docs.

Project Workflow & Responsibility Matrix

Work is assigned in the same order as the names provided by the team.

Milestone	Activities	Assigned Member
1. Model Selection & Architecture	1.1 Research and select the appropriate generative AI model 1.2 Define application architecture 1.3 Set up development environment	**Pavithra M (Team Leader)**
2. Core Functionalities Development	2.1 Develop core functionalities 2.2 Implement FastAPI backend for routing and user-input processing	**Manisha S**
3. routes.py Development	3.1 Write main application logic in routes.py; connect frontend, Gemini functions, and SQLite database	**Monica G**
4. Frontend Development	4.1 Design and develop UI 4.2 Create dynamic templates with FastAPI Jinja2	**Monisha Anandan**
5. Deployment	5.1 Prepare application for local deployment 5.2 Test and verify local deployment	**Mridula M**

FitBuddy System Flow

Step	Process
1	User enters personal details, fitness goal and workout intensity.
2	FastAPI receives and validates the input.
3	Gemini 1.5 Pro generates the personalized 7-day workout plan.
4	Gemini Flash generates a nutrition or recovery tip.
5	User and plan data are stored in SQLite through SQLAlchemy.
6	User can submit feedback; Gemini 1.5 Pro updates the plan.
7	Admin can view users and original/updated plans through /view-all-users.

Source basis: Project description, scenarios, technical architecture, project workflow, and milestone activities from the uploaded FitBuddy project document. The work allocation maps the five milestones to the five team members in the exact order supplied by the team.